

Quantum machine learning resources:

Github repo: awesome QML

<https://github.com/krishnakumarsekar/awesome-quantum-machine-learning>

2 Github repo

<https://github.com/artix41/awesome-quantum-ml>

QML MOOC by Petter Wittek

https://www.youtube.com/watch?v=QtWCmO_Kllg&list=PLmRxgFnClhaMgvot-Xuym_hn69lmzlokg&index=1

QML Hackaton

<https://medium.com/xanaduai/we-came-we-saw-we-ghacked-dacea7914f5a>

QML 101 - Medium post

<https://towardsdatascience.com/quantum-machine-learning-101-1058f24c3487>

The Emerging Role of Quantum Computing in Machine Learning (Recomendado)

<https://www.youtube.com/watch?v=V2WRdU56Wlc>

Introductory material for QML:

<https://quantumcomputing.stackexchange.com/questions/2151/introductory-material-for-quantum-machine-learning>

An introduction to Quantum Machine Learning - Webinar by Aroosa Ijaz

<https://www.youtube.com/watch?v=-DWng3jyBIM>

An introduction to QML by Maria Schuld

<https://arxiv.org/pdf/1409.3097.pdf>

Quantum machine learning for supervised pattern recognition by Maria Schuld:

<https://pdfs.semanticscholar.org/9d3b/04e3667b13d55d0fbe65cb21ea7c52614e2c.pdf>

QML talk 2020: Maria Schuld

https://www.youtube.com/watch?v=C_IBYKV_pJo

QML and PennyLane by Maria Schuld

<https://www.youtube.com/watch?v=pe1d0RyCNxY>

How to translate from Qubiter to PennyLane

https://github.com/artiste-qb-net/qubiter/blob/master/qubiter/jupyter_notebooks/Translating_from_Qubiter_to_Xanadu_PennyLane.ipynb

QML by Jacob Biamonte and Petter Wittek

<https://arxiv.org/pdf/1611.09347.pdf>

“QML Algorithms: Read the Fine Print” by Scott Aaronson

<https://www.scottaaronson.com/papers/qml.pdf>

QML: a classical perspective

<https://arxiv.org/pdf/1707.08561.pdf>

QML for data scientists

<https://arxiv.org/pdf/1804.10068.pdf>

Data re-uploading for a universal quantum classifier

<https://arxiv.org/pdf/1907.02085.pdf>

Talk: What Can We Expect from QML? Dr. Peter Wittek

<https://www.youtube.com/watch?v=EKWGLERVLuc>

Highlighting Quantum Computing for Machine Learning

<https://medium.com/meetech/highlighting-quantum-computing-for-machine-learning-1f1abd41cb59>

Quantum embeddings for machine learning

<https://arxiv.org/pdf/2001.03622.pdf>

On quantum ensembles of quantum classifiers

<https://arxiv.org/pdf/2001.10833.pdf>

Parameterized quantum circuits as machine learning models

<https://iopscience.iop.org/article/10.1088/2058-9565/ab4eb5/meta>

The Holy Grail of Quantum Artificial Intelligence: Major Challenges in Accelerating the Machine Learning Pipeline

<https://arxiv.org/abs/2004.14035>

Quantum Assisted Feed-Forward Neural Network For MNIST Image Classification

<https://quantumzeitgeist.com/quantum-assisted-feed-forward-neural-network-for-mnist-image-classification>

No Free Lunch for QML

<https://scirate.com/arxiv/2003.14103>

End-to-End QML Implemented with Controlled Quantum Dynamics

<https://scirate.com/arxiv/2003.13658>

Bayesian deep learning on quantum computers

<https://arxiv.org/pdf/1806.11463.pdf>

Repo with code of the previous paper:

<https://gitlab.com/apozas/bayesian-dl-quantum>

Latest improved HHL algorithm for previous paper

<https://arxiv.org/pdf/2003.12770.pdf>

QML: Prospects and challenges

<https://www.youtube.com/watch?v=Ycy40s4aXxs>

Panel discussion: QML

https://www.youtube.com/watch?v=-YY_kWpdu3Y

QML with PyTorch and Qiskit

<https://blog.usejournal.com/quantum-machine-learning-hybrid-quantum-classical-machine-learning-with-pytorch-and-qiskit-d03da758d58b>

Reddit question: Best major and focus for QML

https://www.reddit.com/r/QuantumComputing/comments/h9upxv/best_majors_and_focus_for_quantum_machine_learning/

A Classical Algorithm Framework for Dequantizing QML

https://www.youtube.com/watch?v=j4k6aVihQXE&list=RDCMUW1C2xOfXslzPgjXyuhkw9g&start_radio=1

New Directions for Tensor Networks: Machine Learning and Quantum Computing I

<https://www.youtube.com/watch?v=Ql2-doWQYJk>

QML - What Quantum Computing Means to Data Mining (Book) by Peter Wittek

https://www.researchgate.net/profile/Peter_Witte/publication/264825604_Quantum_Machine_Learning_What_Quantum_Computing_Means_to_Data_Mining/links/5ababcfba6fdcc71647085db/Quantum-Machine-Learning-What-Quantum-Computing-Means-to-Data-Mining.pdf

Implementing QSVM on IBM's Quantum computers

<https://quantumcomputinguk.org/tutorials/how-to-implement-qsvm-algorithm-on-ibms-quantum-computers-with-qiskit>

QML: Training a model on IBM quantum computers with the iris flower dataset

<https://iris.entropicalabs.io/>

Introduction to QSVM

<https://medium.com/@patrick.huembeli/introduction-into-quantum-support-vector-machines-727f3ccfa2b4>

Training quantum neural networks with PennyLane, PyTorch, and TensorFlow

<https://medium.com/xanaduai/training-quantum-neural-networks-with-pennylane-pytorch-and-tensorflow-c669108118cc>

How to calculate Quantum Gradients and use them for classification:

https://colab.research.google.com/github/kareem1925/Ismailia-school-of-AI/blob/master/quantum_gradients/universal_qubit_classifier_using_gradient_descent.ipynb

Can Quantum Computers Learn Like Classical Computers? A Co-Design Framework for Machine Learning and Quantum Circuits

<https://scirate.com/arxiv/2006.14815>

On Sampling and Inference using Quantum Algorithms

<https://scirate.com/arxiv/2006.11775>

Recurrent Quantum Neural Networks

<https://scirate.com/arxiv/2006.14619>

Layerwise learning for quantum neural networks

<https://scirate.com/arxiv/2006.14904>

Boltzmann machine learning with a variational quantum algorithm

<https://arxiv.org/abs/2007.00876>

Single-photon QML improves on the classical result

<https://quantumzeitgeist.com/single-photon-quantum-machine-learning-improves-on-classical-result>

The effect of data encoding on the expressive power of variational QML models

<https://arxiv.org/abs/2008.08605>

“Quantum Classification” by Brian N. Siegelwax

<https://link.medium.com/fOEeWHIly9>

“Superdense Encoding of Classical Data” by Brian N. Siegelwax

<https://link.medium.com/ghSSSX2iA9>

“Quantum Clustering & Classification” by Brian N. Siegelwax

<https://link.medium.com/9BTVkL5iA9>